SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME:MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO1*: Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)**

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 5**

Annexure A

Code of Conduct:

I K. Neeha(192425245) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Case Study 1: RL-Based Warehouse Robot

Introduction

A logistics company uses a Reinforcement Learning (RL)-based warehouse robot to automate picking, packing, and transporting items. The robot learns the most efficient routes while avoiding obstacles, reducing delays, and improving overall warehouse productivity.

MDP Components

- **States:** Robot location, package location, obstacle positions, battery level, and destination.
- **Actions:** Move forward, turn left/right, pick an item, pack it, deliver it, or recharge.
- **Rewards:** Positive rewards for successful item pickup and delivery; negative rewards for collisions, incorrect actions, and delays.

- **Policy:** Determines the best action for every warehouse situation to maximize long-term rewards.
- **Environment:** Warehouse containing shelves, packages, obstacles, charging stations, and delivery points.

Reward Function

A balanced reward function ensures the robot performs tasks efficiently:

- **+20** for successful delivery.
- **+10** for picking the correct item.
- **-10** for collision with obstacles.
- **-5** for unnecessary delays.
- **-2** for excessive battery usage.

Evaluation

A well-designed reward function encourages the robot to complete tasks quickly while maintaining safety and accuracy. It minimizes travel time, reduces collisions, and improves warehouse efficiency.

Case Study 2: Streaming Platform Movie Recommendation

Introduction

A streaming platform applies Reinforcement Learning to recommend movies dynamically according to user preferences, viewing history, and feedback. The recommendation system continuously learns from user interactions to improve personalization.

RL Components

- **States:** User watch history, preferred genres, ratings, viewing duration, and search history.
- **Actions:** Recommend a movie, TV show, or a new genre.
- **Rewards:** Positive rewards when users watch, like, or complete recommended content, and negative rewards when recommendations are ignored.

Reward Function

- **+10** for watching a movie completely.
- **+5** for positive ratings or likes.
- **-5** for skipping recommendations.
- **-10** if the user leaves the platform after recommendations.

Exploration vs. Exploitation

- **Exploration:** Recommends new genres or less popular movies to discover new interests.
- **Exploitation:** Recommends content similar to previously liked movies.

Evaluation

A balanced reward function improves user engagement by increasing watch time while maintaining diversity in recommendations through exploration.

Case Study 3: Autonomous Agricultural System

Introduction

An autonomous agricultural system uses Reinforcement Learning to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. The objective is to maximize crop yield while conserving water and fertilizer.

RL Framework

- **States:** Soil moisture, temperature, humidity, weather forecast, crop growth stage, and nutrient levels.
- **Actions:** Increase irrigation, reduce irrigation, apply fertilizer, or take no action.
- **Rewards:** Positive rewards for healthy crop growth and resource conservation, with penalties for wastage and crop damage.

Reward Function

- **+20** for healthy crop growth.
- **+10** for efficient water usage.
- **-10** for over-irrigation.
- **-15** for crop damage or poor fertilizer management.

Delayed Rewards

Crop yield is observed only after several days or weeks, making rewards delayed. The RL agent must learn that current actions influence future harvest quality.

Suitable Learning Algorithm

Deep Q-Network (DQN) or **Q-Learning** is appropriate because these algorithms can learn optimal policies from delayed rewards and continuously improve decision-making.

Evaluation

The RL framework helps reduce water consumption, optimize fertilizer use, increase crop productivity, and support sustainable agriculture.

Case Study 4: Cybersecurity Intrusion Detection

Introduction

A cybersecurity system uses Reinforcement Learning to detect and respond to network attacks in real time. The agent continuously monitors network traffic and learns effective strategies to prevent cyber threats.

RL Framework

- **States:** Network traffic statistics, login attempts, packet information, system status, and detected anomalies.
- **Actions:** Allow traffic, block suspicious IP addresses, isolate infected devices, or notify administrators.
- **Environment:** Enterprise computer network containing normal and malicious traffic.

Reward Function

- **+20** for correctly identifying an attack.
- **+10** for preventing unauthorized access.
- **-20** for missing an attack.
- **-10** for false alarms that block legitimate users.

Challenges

- Cyberattacks occur infrequently, leading to **sparse rewards**.
- The system must make **real-time decisions** without delaying legitimate network traffic.

Evaluation

A properly designed reward function improves intrusion detection accuracy, minimizes false positives, and enhances network security while maintaining normal operations.

Case Study 5: Smart Home Automation System

Introduction

A smart home automation system applies Reinforcement Learning to control lighting, heating, and cooling based on occupancy, weather conditions, and user preferences. The objective is to maximize user comfort while reducing electricity consumption.

RL Framework

- **States:** Room occupancy, indoor temperature, outdoor weather, humidity, and time of day.
- **Actions:** Turn lights on/off, adjust heating or cooling, and control ventilation.
- **Rewards:** Positive rewards for maintaining comfort with minimum energy usage.

Reward Function

- **+15** for maintaining a comfortable indoor environment.
- **+10** for reducing electricity consumption.

- **-10** for causing user discomfort.
- **-5** for excessive energy usage.

Handling Conflicting Objectives

The system balances user comfort and energy efficiency by assigning appropriate weights to both objectives within the reward function.

Evaluation

Reward shaping enables the RL agent to learn faster, reduce electricity bills, improve user comfort, and create an energy-efficient smart home environment.

Conclusion

Reinforcement Learning enables intelligent systems to learn optimal decisions through interaction with their environment. By carefully defining **states, actions, rewards, and policies**, RL can be successfully applied in warehouse automation, recommendation systems, agriculture, cybersecurity, and smart homes. A well-designed reward function is essential for achieving efficient, safe, and reliable decision-making while improving overall system performance.

RUBRICS FOR CASESTUDY

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

		strong justification			
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand